

Blockchain

Uniswap Pair

Address:

0xa14DFBaa23EE8E8b69878381F8Fd1D
0BC502c043

Functions:

```
token0() {return  
"0x9f8f72aa9304c8b593d555f12ef65  
89cc3a579a2";}
```

...

ERC20 Token

Address:

0x9f8f72aa9304c8b593d555f12ef658
9cc3a579a2

Functions:

```
symbol() {return "MKR";}
```

—● : simple retrieval
—◆ : dynamic retrieval

From raw transaction to contract-level semantic

Uniswap event log

Address 0xa14DFBaa23EE8E8b69878381F8Fd1D0BC502c043

Name TokenPurchase (address buyer, uint256
eth_sold, uint256 tokens_bought)

Topics

[0]: 0x31e085afd48a1d6e51cc193153d625e8f0514c7f
[1]: 56376237623762376237623786
[2]: 5134557581

Code: emit TokenPurchase (buyer, eth_sold,
tokens_bought)

// Uniswap Model

```
if (EventLog.Name == "TokenPurchase": {  
    semantic.action = "Swap";
```

```
    semantic.from = EventLog.Topics[0];  
    semantic.inToken = "ETH";
```

// query token symbol to Blockchain

```
let erc20 = chain(EventLog.Address).token0();  
semantic.outToken = chain(erc20).symbol();
```

```
    semantic.amountIn = EventLog.Topics[1];  
    semantic.amountOut = EventLog.Topics[2];
```

```
}
```

```
else if (EventLog.Name == "Swap": {  
    semantic.action = "Swap"; ...
```

```
}
```

① getERC20 Addr

② token symbol

emit